5 Plant Reproduction

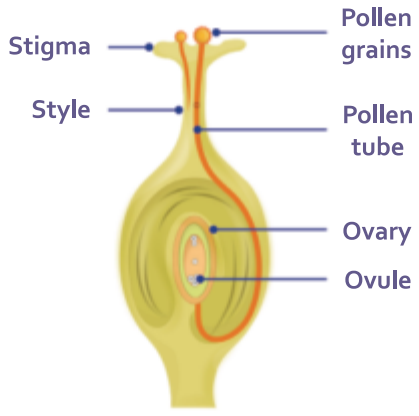
The process by which a living organism produces more of its own kind is called reproduction. Plants reproduce through seeds, stems, roots, or leaves. We will learn about the different types of plant reproduction and ways of seed dispersal in this chapter.

Learn about

- Sexual reproduction in plants: Fertilization
- Structure of a seed
- Germination
- Dispersal of seeds
- Vegetative reproduction in plants

Sexual Reproduction in Plants: Fertilization

When a pollen grain from the anther of a flower falls on the stigma of the same kind of flower, it may germinate because of the presence of a sticky substance released by the stigma. The pollen grain then sends out a tiny thread-like tube called a pollen tube. It carries the male reproductive cell (or gamete) through the style of the pistil to the ovules present in the ovary. The male gamete then enters an ovule and combines with the female gamete (or egg) to form a fertilized cell called the **zygote**.



Fertilization

*The process in which the male and female gametes fuse together to form a zygote is called **fertilization**.*

Fact File

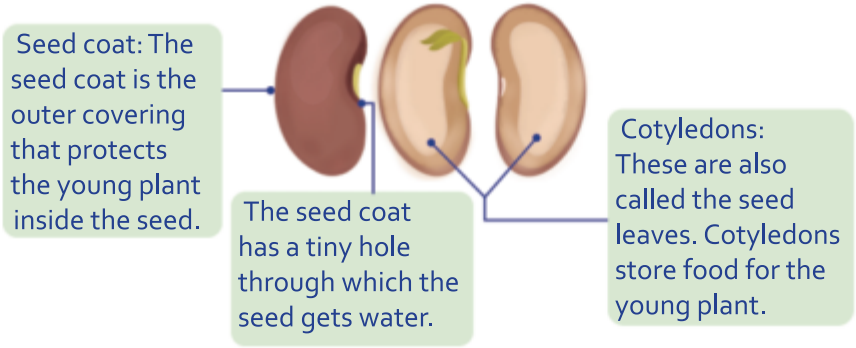
Understanding the difference between pollination and fertilization can be tricky because they both involve the transfer of something (pollen) in plants to make seeds. Pollination is the movement and deposition of pollen grains on the stigma. It is an external process that does not directly involve the fusion of gametes. Fertilization is the internal process within the ovule where the male gamete from the pollen grain fuses with the female egg, leading to zygote formation.

Formation of Fruit and Seed

After fertilization, all the parts of the flower, except the ovary, become dry and fall off. The zygote (present inside the ovule) after many divisions, develops into an embryo. The embryo further develops to form the baby plant. The fertilized ovule forms the seed, and the ovary becomes the fruit.

▶ Structure of a Seed

A seed has an outer covering called the seed coat. The seed coat has a tiny hole through which the seed gets water. Seeds usually have one or two seed leaves known as **cotyledons**. When a seed has two cotyledons, they are generally thick and fleshy structures, as they store food inside them. Seeds with only one cotyledon do not store food inside the cotyledon. In such seeds, food is stored in some other part of the seed.



Structure of a bean seed

▶ Germination

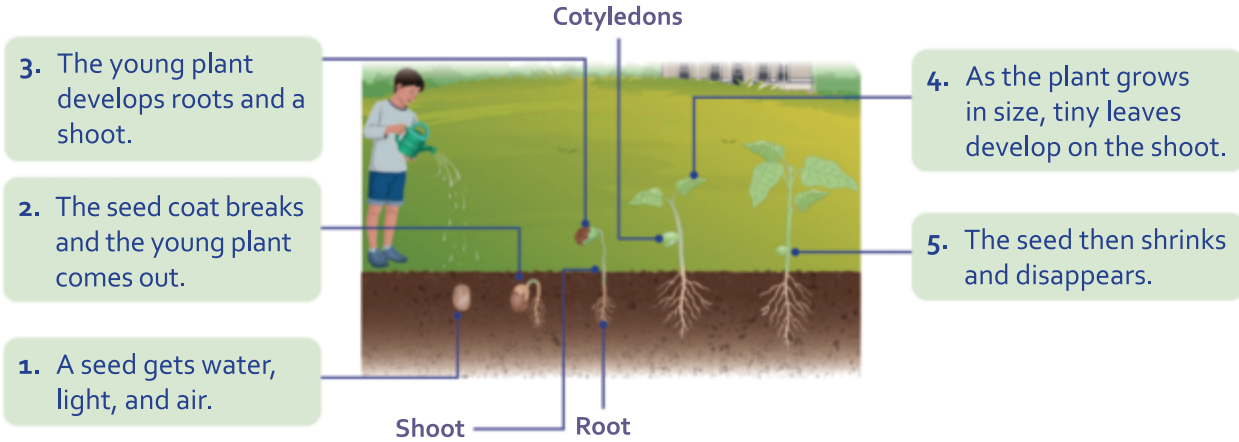
The process by which a seed grows into a young plant, or seedling, is called ***germination***. A seed germinates only when it gets enough light, air, water, and warmth. During the early stages of germination, the seedling uses the food stored inside the seeds for its growth.

When the seed receives adequate light, water, air, and warmth, it becomes soft and spongy. The seed coat then breaks and the young plant comes out. The young plant develops roots and a shoot. Tiny leaves develop on the shoot and start making food. This young plant is called a **seedling**. The seedling absorbs water and nutrients from the soil with the help of its roots. As the plant grows in size, it develops more leaves. The seed then shrinks and disappears. The seedling grows to become a new plant.

Fact File

Some seeds need cold temperatures for their seed coat to break their inactive state. This is called **stratification**.

All seeds do not grow into new plants because some are eaten by animals, some get destroyed by rain or wind, while some get insufficient air, water, or warmth.



Stages of germination of a bean seed

Activity

Aim: To study the conditions necessary for the germination of seeds

Materials required: A glass slide, an adhesive tape, some bean seeds, a beaker, and water

Procedure:

1. Take a glass slide. Fix three bean seeds on it as shown in Fig. A. Label the seeds as Seed 1, Seed 2, and Seed 3.
2. Place the slide in a beaker. Pour water in the beaker till Seed 1 is completely under water; Seed 2 is half below and half above the level of water; and Seed 3 is completely above the level of water (Fig. B).

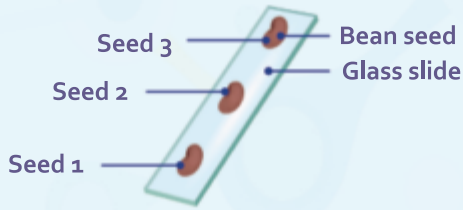


Fig. A

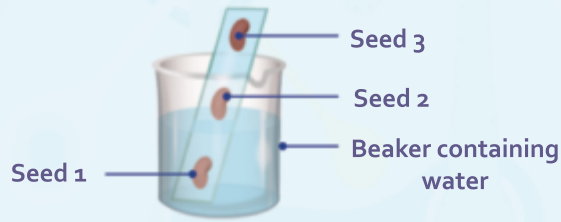


Fig. B

3. Keep this beaker in a warm place for 2–3 days (Fig. C).

Fig. C

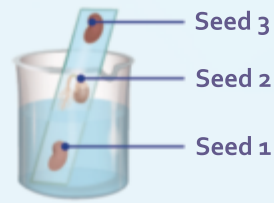


Fig. D

Observation: Seed 1 grows in size but does not germinate. Seed 2 develops root and shoot (Fig. D). Seed 3 does not germinate either.

Conclusion: Seed 1 does not germinate because of the absence of air. Seed 2 germinates since it gets both water and air. Seed 3 receives air, but does not get water for germination. Therefore, air, water, and warmth are needed for germination.

Questions

Write *T* for True and *F* for False.

1. After fertilization, the seed becomes the ovule. .....
2. The pollen grains sends out a tiny thread-like tube called a pollen tube. .....
3. The male and female gamete fuse together to form the cotyledon. .....
4. The seed needs air, water, light, and warmth to germinate. .....

Dispersal of Seeds

*The process by which the seeds are scattered away from the parent plant is called **seed dispersal**.*

Need for Seed Dispersal

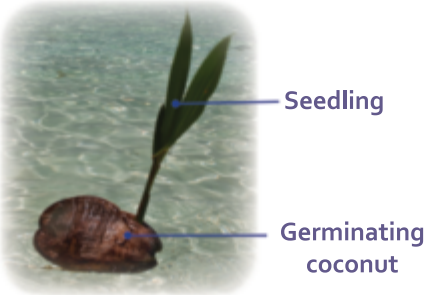
Seed dispersal is important for the proper growth and survival of the parent plant as well as new plants. If seeds do not get dispersed far away, they will grow very close to the parent plant. They will then compete for light, air, space to grow, water, and nutrients in the soil. Many of them would die. Hence, many plants have developed ways to scatter their seeds away from them. This ensures that the seeds do not grow too close to one another. There are different ways by which seeds are dispersed.

Dispersal by Wind

Seeds of some plants are light in weight and have wing-like structures. They are easily carried away to other places by wind.

- **Dandelion seeds** have hair that act like a parachute.
- **Maple seeds** have wing-like structures. The wind carries these seeds away from the parent plant.

Dandelion flower dispersing its seeds



Seeds of the coconut palm are dispersed by water.

Dispersal by Water

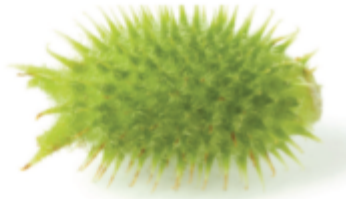
Seeds of plants that grow in or near water are dispersed by water. Fruits of the coconut palm have a fibrous covering that makes them lighter, and helps them to float on water.

Dispersal by Animals

Animals and human beings eat fruits and throw away seeds, which grow into new plants.

Squirrel eating nuts

- Some animals and birds eat fruits, berries, and nuts, along with their seeds. The seeds pass through their bodies undigested and are dispersed to other areas.
- Some seeds have hooks, stiff hair, or spines. The hooks of cocklebur plant are an example. These seeds stick to the bodies of animals and birds and are carried away to different places.



Hooks of cocklebur

Pea plant dispersing seeds

Dispersal by Explosion of Fruit

Some fruits, such as the pea and balsam, burst open forcefully when they dry and liberate their seeds. The seeds fall on the ground away from the place where the parent plant is growing.

REMEMBERING

Questions

List the ways in which the seeds of the given plants are dispersed.

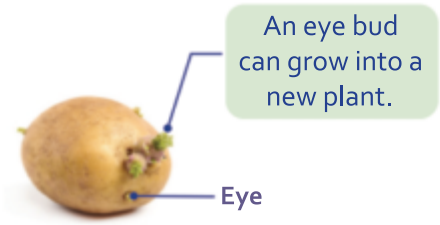
1. Pea seeds .....
2. Dandelion seeds .....
3. Coconut fruit .....
4. Maple seeds .....
5. Balsam seeds .....
6. Cocklebur seeds .....

Other Ways of Reproduction in Plants: Vegetative Reproduction

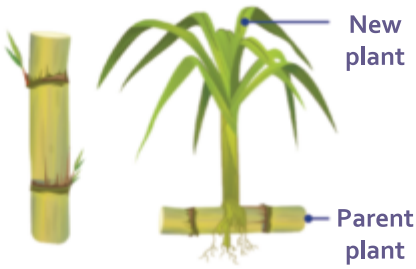
Vegetation reproduction *or* ***vegetation propagation*** *is the process in which new plants grow not only from seeds, but also from other parts such as stems, leaves, and roots in certain plants.*

Stems

We eat the stems of potato, ginger, onion, and sugarcane. The stems of these plants can grow into new plants. A potato is a special stem that grows under the ground. We may find special structures called **'eyes'** on its surface. Any piece of potato with an eye bud can grow into a new plant.

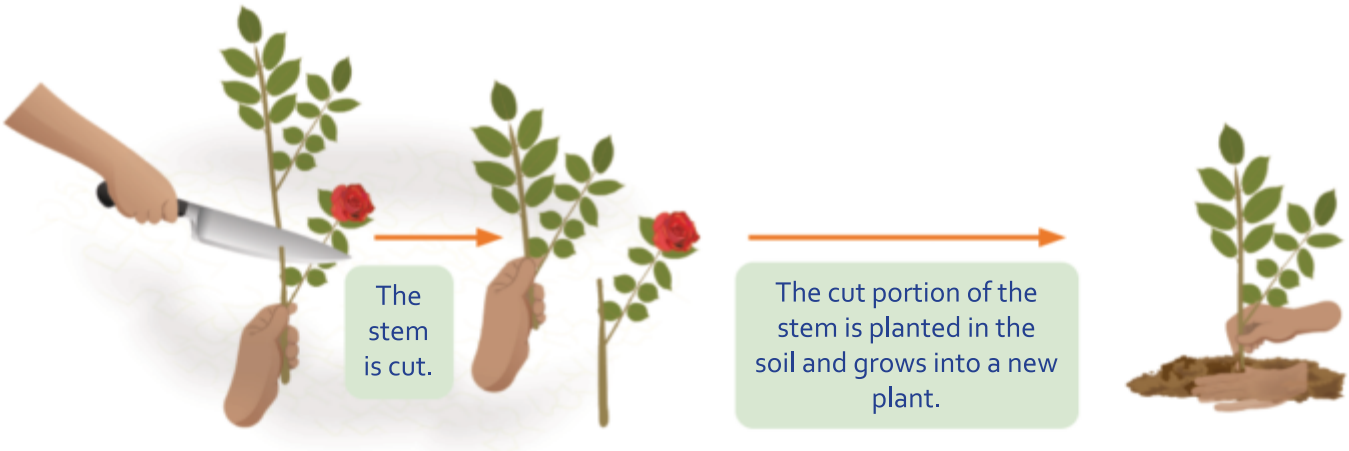


Potato stem giving rise to a new plant



Sugarcane stem giving rise to a new plant

The stems of plants such as the rose, *Hibiscus*, and money plant, too, can be used to grow new plants. In ginger, the stem has small structures called buds. A cutting of such a stem along with the bud can grow under the ground into a new plant.

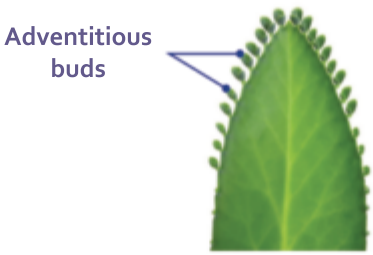


The stem of a rose can grow into a new plant.

An onion has a special structure called a bulb. The new plant can develop from the bulb.

Leaves

The leaves of *Bryophyllum* have several buds along their margins. New plants grow from these leaf buds when they fall onto moist soil.



Leaf buds of *Bryophyllum* can grow into new plants.

The sweet potato is a root that can grow into new plants

Roots

The roots of plants such as the carrot, turnip, and sweet potato grow to develop shoots when kept in water for some time. These shoots later grow into new plants. Roots of carrot are tuberous.

When planted in specially prepared soil, they develop buds, which grow into leafy shoots. These shoots are then detached from the parent plant and grown separately.

Activity

Aim: To grow new plants from leaves and stem

Materials required: Soil, manure, coco peat, a small pot or a paper cup, *Bryophyllum* leaves, pieces of potato with buds known as 'eyes', and a fresh turmeric piece (can be bought from a vegetable seller during winter), and water

Procedure:

1. Mix soil, manure, and coco peat in a small pot or a paper cup.
2. Place the *Bryophyllum* leaf and keep it over the soil mix. Water it whenever required.
3. Observe after 15–20 days.

Observation: You will see shoot emerging from the margin of the leaf. Repeat this activity with the potato piece and turmeric piece too. Place them beneath the soil.

Activity

Aim: To study reproduction through roots

Materials required: A carrot, a shallow dish, and water

Procedure:

1. Cut off the top portion of a carrot root.
2. Place the cut top in a shallow dish with the cut end facing downwards.
3. Pour water into the dish such that half the portion of the carrot remains dipped under water.
4. Keep the dish in a place where the carrot top receives enough sunlight.

Observation: After a few days, a new plant will grow from the carrot top.

Conclusion: Some plants, such as carrots, reproduce through their roots.

Spores

Some plants do not bear flowers and seeds. They bear special structures, called **spores**, for reproduction. Mosses and ferns reproduce through spores.

Mosses bear spores

Questions

Fill in the blanks.

1. Any piece of potato with an ..... can grow into a new plant.
2. The ..... of *Hibiscus* plant can be used to grow new plant.
3. The leaves of *Bryophyllum* form several buds along its ..... that can grow into a new plant.
4. Mosses reproduce through .....

Wrap Up

- The fusion of the male and female gametes to form a zygote is called fertilization.
- The fertilized ovule becomes the seed and the ovary becomes the fruit.
- The process by which a seed grows into a young plant, or seedling, is called germination.
- The process by which the seeds are scattered away from the parent plant is called dispersal.
- Seed dispersal is important for the proper growth and survival of the parent plant as well as the new plants. There are different ways by which seeds are dispersed: by wind, water, animals, and explosion of fruits.
- New plants grow not only from seeds, but also from other parts such as stems, leaves, and roots in certain plants. This is known as vegetative reproduction.

Exercises

SECTION I

A **Choose the correct option.**

1. The pollen grains are released from ..... .
   a. cotyledon b. anther c. filament d. style
2. Pollen grain germinates and form ..... .
   a. style tube b. seed coats c. pollen tube d. cotyledons
3. Which term describes the fertilized cell that will eventually develop into a new plant?
   a. Zygote b. Ovule c. Pollen d. Female gamete

4. After germination, a young plant emerges from the seed. This young plant with its first true leaves is known as a/an ..... .
   a. seedling b. ovule c. cotyledon d. pollen
5. Dandelion seeds have a unique adaptation that allows them to travel long distances. They possess a parachute-like structure that facilitates dispersal by ..... .
   a. water b. wind c. animals d. explosion

B Assertion and Reasoning questions

1. **Assertion (A):** Fertilization is the process in which pollen grains land on the stigma of a flower.

   Reason (R): Fertilization leads to the formation of a zygote, which is the first stage of a new plant.
   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True
2. **Assertion (A):** Animals can help disperse seeds.

   Reason (R): Some seeds have hooks or spines that can attach to animals.
   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True

C Choose the correct option to fill in the blank.

1. A seed has an outer covering called the ..... (seed cover/ seed coat).
2. The pollen grain may germinate because of the presence of a ..... (dry/sticky) substance released by the stigma.
3. The zygote develops into a/an ..... (embryo/cotyledon) after fertilization.
4. Dandelion seeds have hair-like structure that acts like a ..... (wing/parachute).
5. Dispersal by explosion takes place in ..... (balsam/maple) fruit.

D Give two examples of the following.

1. Seeds dispersed by water ....
2. Seeds dispersed by animals ....
3. Plants that can grow from stems ....
4. Plants that can grow from roots ....
5. Plants that reproduce through spores ....

SECTION II

E Short answer questions

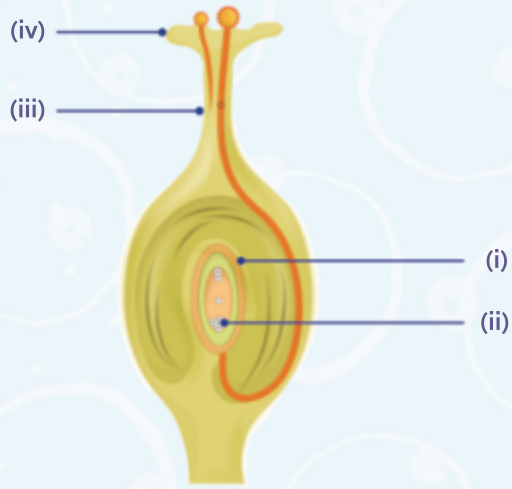
1. Define fertilization in case of plants.
2. What is a seed coat? What is the role of the tiny hole that is present on the seed coat?
3. When a seed has two cotyledons, why are they thick and fleshy?
4. What do you understand by germination of a seed?
5. A farmer's corn seeds struggle to sprout despite good weather. Why do all seeds not grow into new plants?
6. Which part of a rose plant can be used to grow new plants?

F Long answer questions

1. How does formation of fruits and seeds take place after fertilization?
2. Describe the structure of a bean seed.
3. Why is there a need for seed dispersal?
4. Explain with examples how seed dispersal by wind takes place.
5. Describe other ways of reproduction in plants apart from seeds.

Picture Study

Look at the picture given below and answer the following questions.



1. What does the given picture show?

   .....

2. Label the parts (i), (ii), (iii), and (iv).

.....

3. On germination, what comes out from the pollen grain?

.....

4. Where does it pass through? What does it contain?

.....

.....

5. What is formed after the fusion of male and female gamete? Where does it occur and what is this process called?

.....

.....

.....

My Learning Corner

SCIENTIFIC SKILLS · ANALYSING · EVALUATING · CREATING

A Think about

VALUES

1. Nutan was helping her mother in her kitchen garden. Suddenly, she noticed a *Bryophyllum* leaf (with small plantlets at its margin) that had detached from the parent plant, was lying on the soil. Small plantlets were also growing in the soil near it. She then asked her mother, "Ma, why is it that the new plants can grow from the *Bryophyllum* leaf, but not from the rose leaf?" Can you answer her question?

.....

B Try out TP

ART INTEGRATION

1. You have studied the male and female reproductive parts of the flower in the previous chapter. Their figures are given below. Draw a similar figure in the space provided, and label the different parts.

2. Visit a nearby nursery with your parents. Converse with the gardener and know the various ways in which he propagates new plants, apart from the conventional method—by seeds. Click pictures of them, print, and paste them in your Science scrapbook.
3. Collect common seeds of different kinds and sizes. Find out their names from elders. Classify them into groups based on their dispersal method. Take the help of Internet for it.
4. Visit the herbal garden of your school. Select a suitable space for growing potato through the eyes present on its surface. A small piece of potato with 2–3 eyes can be used for propagation. You can perform this activity individually or in groups.
5. Potato is an important food crop of the world. You have studied in this chapter how potatoes are cultivated by vegetative propagation. It is grown nearly in all states of India, but there are some states that are major producers of potatoes. On a political map of India, mark these states. Also, find out from the Internet, the soil and climate needed to grow potatoes.

Self-Assessment

Now that you have completed the chapter, score each of the following tasks from 1 to 5 to indicate how well you can do them.

Score 5 = I can definitely do this.

Score 1 = I cannot do this yet.

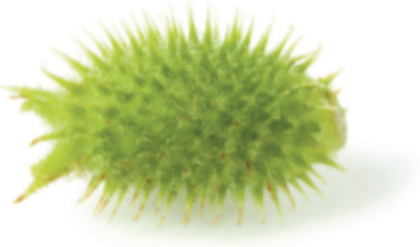
I can...	My score
• draw and label the male and female reproductive parts of a flower.	
• describe the need for the process of fertilization in plants.	
• explain the process of fertilization in plants.	
• discuss the significance of seed dispersal and give examples of each kind of seed dispersal.	
• identify various parts through which vegetative reproduction takes place and give examples.	

Worksheet

Look at the pictures and complete the sentences given below.

1. This is a .....
2. Its fruits has a ..... covering.
3. It helps it to .....
4. Dispersal of coconut thus takes place by

   .....



1. This is a ..... plant.
2. The seeds of cocklebur plant have .....
3. These seeds thus sticks to the .....
4. So the dispersal of seeds takes place by .....

1. This is a figure of .....
2. It shows vegetative propagation through .....
3. Other plants that also show vegetative propagation through roots are:

a. .....

.....

b. .....

.....

c. .....

.....